



Figure 26 The Kahuterawa Stream near Scotts Rd. Good pasture cannot be maintained on steep slopes. Invasion by bracken fern begins the return to forest. Browntop, Chewing's fescue, and *Notodanthonia* predominate on the less steep parts. 1964

Coprosma robusta, and *rangiora*. In the absence of opossums five-finger would become an important part of the vegetation but this is usually destroyed as it emerges above the bracken. *Dicksonia squarrosa* follows soon after, then *hinau*, *rewarewa*, *Cyathea medullaris* and, in some places, *kamahi* but the seed source of this species is now very limited. At this stage or earlier, bracken disappears and *Coprosma robusta*, *rangiora*, *kaikomako*, and *putaputaweta* persist as long as they can keep some of their foliage in the canopy. *Tawa*, as is characteristic of the species, makes a late entry and it is about this time that *nikau*, more tree ferns, and many of the ground ferns make an appearance. In the Kahuterawa valley there are some magnificent dense stands of *Cyathea medullaris* (figure 27). In other places *tawa*, *kamahi*, or *mahoe* have assumed dominance.

Regeneration through manuka

On the terraces manuka is more abundant than bracken. One small area in the Kahuterawa valley was studied by Grant (1966). Other areas at Aokautere were used to provide information about the dynamics of manuka stands (Esler, unpublished). The results are presented here.

Manuka stands are most extensive on the strongly leached soils formerly occupied by black beech forest in the Aokautere district. The shrubland on

Figure 27 Mamaku in the region of Butlers Creek. Half a century of regeneration has produced elegant stands of mamaku along the streams in the hill country. 1971



these soils adjoining Forest Hill Rd is a good example of the fact that the richest floras are not on the richest soils. In the Manawatu, as in many other districts, fertile soils quickly become occupied by alien plants once the original vegetation has been disturbed, and few natives can compete against them. The reduction in plant diversity is taken a step further when the few dominant pioneers suppress the many colonists once the colonising phase has passed. Because the poor soils of the Aokautere district do not readily support good pasture many weedy species and native plants gain a foothold. The reversion to manuka is rapid unless pastures are well managed. Shrubland which has developed from inferior pasture contains a diversity of species, both native and introduced. In fact, the shrubland has two distinct floras—the first in the transitional phase from pasture to scrub and the second in the more mature shrubland which contains the embryo of a forest. The success or failure of the outgoing and the incoming species is regulated to a degree by the changes in light intensity on the ground as the manuka passes through its cycle from establishment to old age. Some details of the phases have been worked out in these Aokautere manuka stands (figure 28).

Manuka establishes from seed on fairly denuded but stable soil. The form of the manuka plants varies with their density. Plants invading pasture are widely spaced and have spreading branches while those establishing among them later are more erect. The older plants are unable to assume a more erect habit quickly enough to compete with the younger members and are soon replaced. When manuka establishes after a fire there is usually a very

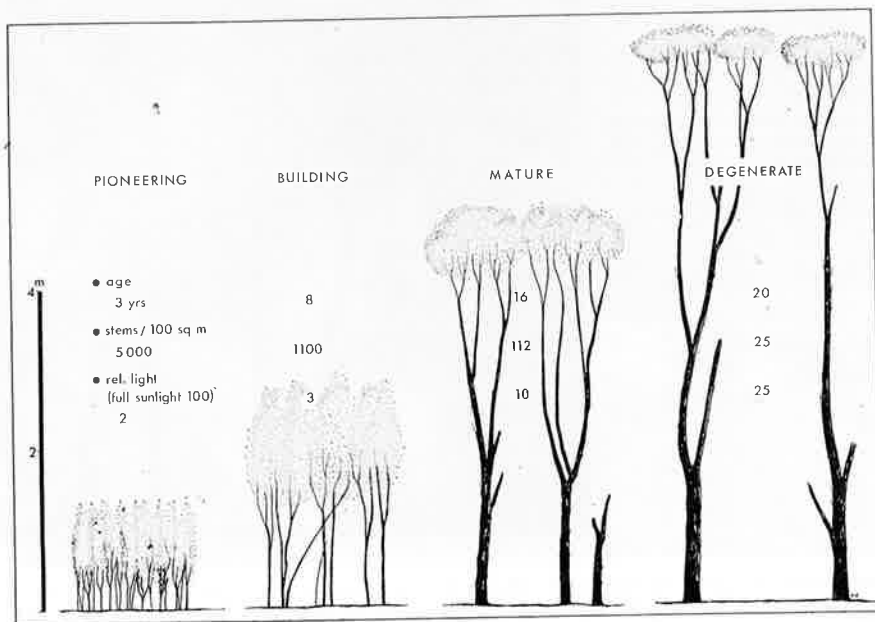


Figure 28 Changing conditions in manuka stands in the Aokautere district related to pattern and process in the plant community.

high density of even-aged seedlings. By either means a high stem density develops but in time intraspecific competition regulates the number of stems. No seedlings of manuka or of shade-tolerant species establish under the closed canopy and a nearly pure manuka stand develops. During this phase the plants grow taller, and although stem density decreases through self-thinning, the amount of light reaching the bases of the plants does not change greatly because dead stems and dead twigs restrict light penetration.

Later many stems weakened by stem borer break off and most of the lower branches fall and decay. At the same time the canopy opens and decreases a little in depth. More light is admitted through the canopy and through the lower layers of the stand, and at the margins of each crown. A greater proportion of the solar energy arrives at ground level as direct sunlight.

By the time the stand has reached old age the canopy has become flat-topped and shallow, diminishing gradually as plants die back branch by branch. The stage is then set for other species to occupy the canopy.

In his classic paper "Pattern and process in the plant community", Watt (1947) demonstrated how performance of dominants is affected by their age. He recognised four stages—pioneering, building, mature, and degenerate. In the pioneering and building stages plants are consolidating their position in

the community, and in the mature and degenerate stages they weaken and offer little competitive resistance to invaders. Stands of manuka follow the same sequence.

At Aokautere the building stage in young manuka shrubland excludes the plants which were constituents of the degraded pasture—*Schoenus apogon*, *Carex flacca*, *Microlaena stipoides*, *Nertera setulosa*, and *Acaena anserinifolia*. Establishment of manuka ceases at this point. At the mature stage the canopy of manuka opens sufficiently to enable a new flora of shade-tolerant species to establish. Among the shrubs which make an appearance at this time are mahoe, rangiora, heketera, *Geniostoma*, lancewood, and *Coprosma* spp. In successions of this nature most vines except supplejack make an entry following the decline of the woody pioneer. At Aokautere the vines most frequently encountered are *Metrosideros diffusa*, *M. perforata*, *Rubus australis*, *R. cissoides*, *Clematis paniculata*, *C. foetida*, and *Parsonsia heterophylla*. Ferns appear in abundance but only *Cyathea medullaris* forms part of the canopy. Sedges of the genera *Carex* and *Uncinia* are a typical feature.

Regeneration of black beech through manuka has been observed in a few places near beech stands.

Except in protected areas the opening of the manuka stand readmits grazing animals which prevent the development of a low mahoe-dominant forest and maintain a shrubland of unpalatable species in which *Cyathodes fasciculata* and *Coprosma rhamnoides* are conspicuous.

Gorse

Gorse is troublesome east of the Manawatu River at an altitude of around 245 m. Control is difficult on the steep slopes where good pasture cannot be maintained. On the easier slopes the paddocks are often too large to enable the concentration of stock which must follow herbicide treatment. Similar problems are encountered on some of the steeper faces on the downlands but the stands are less extensive there. In a few places in the Manawatu native trees and shrubs have established in gorse and are replacing it.

Pepper tree

Pepper tree areas are confined to the higher parts of the Manawatu, the largest stands being around Tarakamuku, Sheres trig and Scotts Rd. The unpalatability of the seedling gives it a considerable competitive advantage in grazed areas.

Tree lupin

Tree lupin is mostly on the sand country but there is some along the courses of the Rangitikei, Manawatu and Oroua Rivers, and the Mangaone Stream.

Near the coast seed is sown on sand already partly stabilised by marram in preparation for planting of trees. Here it serves a useful purpose in fixing sand and providing nitrogen. It also occupies some of the older sand plains where reasonable pasture can be maintained. In these places it is regarded as a nuisance and the noxious weeds inspector for the Manawatu County uses his discretion in urging reasonable control of this anomalous plant which is useful in one situation and troublesome in another.

GRASSLANDS

There were grasslands in the Manawatu in Maori times but we know nothing of their composition. A small patch of natural grassland may have existed at the highest point in the district but it is certain that grassland was extensive on parts of the lowlands. Over 90% of the district is now grassland, mostly pasture of some sort. In the water reserves grassland has developed as the end point of forest degeneration.

Early grasslands on the lowlands

The greatest extent of grassland in the early days was on the sand country. The vegetation of the coastal strip undoubtedly fell into the category termed "sand grass dune" by Cockayne (1911). *Spinifex* was the dominant grass and it seems likely that there was some *Deyeuxia billardieri* as well. The only other conspicuous monocotyledon would have been the sedge *Desmoschoenus*. Marram was not introduced to the Manawatu coast until about the 1890s.

Grasslands away from the coastal strip are mentioned in at least three early accounts. Wakefield (1845, volume 2, page 234) described his trip from Parewanui (then a kilometre nearer the Rangitikei River than the present settlement with that name) to the Oroua River. It appears that in skirting the terrace country he passed near the site of Carnarvon, crossing sand and alluvium. He wrote in his diary in 1840: "We travelled all day through open pasture land, the path apparently avoiding the timber parts . . .". Richard Taylor, the missionary, followed a similar route in 1849 (Taylor 1833-73) and commented on the level grass plain which extended as far as the eye could see.

A statement by Buick (1903, page 286) indicates extensive grassland in the southern stretch of the sand country also: "Between Foxton and what is now known as Oroua Bridge the predominating feature of the landscape was a succession of sand-hills, which rose in long undulating ridges, like great billows of the sea. With rude regularity these grass-covered dunes pointed east and west, as indicating the direction of the winds which formed them.

Here and there over the wide extent of country which they covered there grew in great abundance the manuka, toetoe, and tussock, while between the sandy ridges there would sometimes nestle acres of rich soil, or perhaps a swampy marsh fringed with green rushes and raupo . . .".

Much of the grassy sand country mentioned has been stable for many centuries and at least some of it is well supplied with moisture. The question of why this land supported grassland rather than scrub is unanswered. A second problem arises—that of the botanical composition of the grassland. Undoubtedly there was toetoe in some places but the impression of short grassland is given by the descriptions. By associating available species with their known habitats we may expect the physiognomic species to have been *Microlaena stipoides*, *Notodanthonia* spp., and *Poa anceps*—but this is pure conjecture. Other low plants which may have been present are *Geranium* spp., *Acaena* spp., *Carex* spp., *Pimelea prostrata*, *Scirpus nodosus*, and *Muehlenbeckia complexa*.

Present-day grasslands

There is at least one pasture type for each soil type in the Manawatu but three broad groups are recognised on the basis of their botanical composition.

(1) Pastures of the dry sand country. These are characterised by the absence of ryegrasses and by the abundance of annual grasses and annual clovers.

(2) The ryegrass-white clover pastures with a predominance of ryegrasses and white clover but containing considerable quantities of perennial weeds, particularly flatweeds and docks.

(3) Pastures on the leached soils of the terraces and hills which lack ryegrasses and are dominated by browntop, Chewings fescue, and *Notodanthonia clavata*. They also have conspicuous amounts of native herbs and weedy plants which do not occur in other pastures.

Pastures of the dry sand country

Pastures facing a period each summer when there is insufficient moisture to allow continuous growth cover the dunes and the dry sand plains. They occur also, to a minor degree, on the light soils of the river margins and where rock or gravel in the hill country is only thinly covered by soil.

On the recently-stabilised dunes close to the coast pasture of a sort develops among the marram tussocks which decline in vigour as the sand movement ceases. There are only a few centimetres of topsoil to hold moisture and the period of active growth spans a few months in late autumn, winter, and early spring when soil moisture is not rapidly dissipated in evaporation. The species are mostly annuals but there are a number of deep-rooted perennials. *Bromus diandrus*, *Vulpia bromoides*, and haretail set seed and die off in the late

spring. Along with *Vulpia myuros*, which is rather less abundant, these and other winter annuals have an important role in providing a quick cover to eroding sand which is temporarily still at a season when it is wet and winds are less severe. *Deyeuxia billardieri* persists through the summer but is not dense enough to provide much erosion control. The legumes—black medick, King Island mellilot, and haresfoot trefoil—are widespread, but there are many apparently suitable sites where they do not occur. Dove's foot is nearly always present in small amounts but *Erigeron floribundus*, Scotch thistle, Californian thistle, and sow thistle are more patchy in their distribution. There are very few places where catsear and hawkbit do not occur. Cocksfoot and tall fescue are quite important pasture constituents. It is usual to regard tall fescue as a plant of moist places where it is more robust and therefore more conspicuous, but it is tolerant of very dry conditions where grazing is not severe. Tree lupin is about the only tall plant in these situations but away from the coast bracken is abundant in some places. *Poa pratensis* and sheep's sorrel have been noted in a few places near the coast but cannot be regarded as typical inhabitants. *Notodanthonia clavata* has been seen within a kilometre of the coast at Himatangi but its presence there is unusual.

Dunes and dry plains of the second dune phase (figure 29) and the dunes of the third are also subject to drought, and dryland species also predominate here. These pastures are also more intensively grazed than the recently-stabilised dunes and there is less refuge for *Erigeron floribundus*, sow thistle, and tall fescue. The more open habitat allows the growth of such annuals as parsley piert, allseed, small-flowered buttercup, *Poa annua*, and *Montia verna*. The major grasses are browntop, cocksfoot, sweet vernal, *Vulpia bromoides*, *Notodanthonia* spp., and *Microlaena stipoides*, with barley



Figure 29 Pastures containing mainly annual grasses and annual clovers on the dunes of the second phase. Tree lupin is in the foreground. Cabbage trees are more plentiful on the plains than on the dunes. 1971

grass, *Aira caryophyllaea*, and *Eragrostis brownii* prominent in some places. Subterranean clover is the main pasture legume but white clover and suckling clover are there in small amounts. Marram has been planted in some places where blow-outs have occurred and persists for a short time after the sand has become stabilised.

On these light soils where plants have an enhanced value as soil stabilisers as well as being producers of herbage, the concept of a weed is not clear-cut. However, some plants are undesirable because they cause obstruction or take the place of more useful species. In this class fall the thistles (mainly Scotch, Californian, and winged), tree lupin (which is useful only when building up nitrogen in the early stages of stabilisation), gorse, and bracken. *Scirpus nodosus* is fairly common and its presence leads many non-botanists to believe that rushes grow on dry dunes. However, *Juncus australis* does grow on comparatively dry sand plains in places too arid for any of the other species of rush.

Ryegrass – white clover pastures

The "improved" pastures are the high-producing grasslands for which the Manawatu is well known. They grow mostly on the alluvial plains but can be maintained on the moist sand plains, parts of the terrace country, and in pockets in the greywacke hills. Such pastures are frequently regarded as pure pastures but any one hectare is likely to contain up to 20 species in addition to the perennial ryegrass, ryegrass hybrids, cocksfoot, timothy, white clover, and red clover which are sown in the seed mixtures. Few pastures are without dandelion, catsear, hawkbit, *Bellis perennis*, narrow-leaved plantain, broad-leaved dock, clustered dock, *Veronica serpyllifolia*, field madder, annual mouse-ear chickweed, suckling clover, *Poa annua*, *Poa trivialis*, Yorkshire fog, sweet vernal, crested dogstail, and *Thuidium furfurosus*, the ubiquitous golden moss. In total more than 40 weedy species regularly occur in high-producing pasture, and this does not include the many annual plants which appear when pasture is damaged.

The relatively light infestation of thistles is a reflection of the potency of herbicides; Scotch, Californian, winged, and slender winged thistles are the most troublesome. Some farmers on the downlands claim that Scotch thistle is a nuisance in new pastures but the time pastures are due for renewal the fertility has declined to such a degree that winged thistle is of greater significance. Nodding thistle is not a problem even though it has been in the district for many years. Marsh thistle is abundant in the Manawatu but is never of any consequence where ryegrass and white clover are the dominants.

Variation in habitat brings variations in the weed floras. The pastures of wet areas have quite a different botanical composition from those of well drained sites. Differences from place to place are quite marked where there are irregularities in topography.

Wetness promotes an abundance of creeping buttercup (less commonly giant buttercup which is significant only in the Linton district), *Agrostis stolonifera*, clustered dock, pennyroyal, *Lotus pedunculatus*, *Juncus articulatus*, and the tall pasture rushes. Although rushes are a conspicuous feature of the wet pastures only a few of the 21 species recorded in the Manawatu are abundant in this habitat. *J. effusus*, an introduced species, occurs in many diverse communities from near the coast to the summit of the range. It usually grows where the water table is constantly fairly high. Being soft-stemmed it is frequently closely grazed but in spite of its palatability and susceptibility to herbicides it continues to be an important species. In its habitat range it approaches *J. gregiflorus* but *J. gregiflorus* can occupy sites which appear to be less wet. There is a temptation to regard *J. effusus* as "the common pasture-inhabiting rush" but this cannot be stated without qualification as there are many depressions where *J. sarophorus* is just as abundant and places on better-drained soils where plants of *J. australis* outnumber all other rushes. The association of rushes with wet soils is strikingly demonstrated by the occasional occurrence of rushes growing in narrow strips about 20 m apart where repeated ploughing in lands has left parallel channels across a paddock.

Droughty patches occur in some ryegrass-white clover pastures on coarse-textured alluvium. These normally remain green throughout the year but in a drought season musky storksbill and *Vulpia bromoides* usually invade.

Small differences in microtopography promote large floristic differences on grazed slopes where sheep tracks follow the contours. Above each path there is a gentle slope surmounted by a minor bank on the edge of the path above. These are three distinct topographic facets. The paths gather most of the dung, urine, and rain wash, and are subjected to the most severe trampling. This favours ryegrass and white clover. The slopes are dominated by sweet vernal and hawkbit while the bank immediately below each path supports browntop, or *Notodanthonia clavata*, or a mixture of the two (see Rumball and Esler 1968). Such mosaics break down any attempts to classify pastures in any rigid fashion on a floristic basis.

Pastures on the leached soils of the terraces and hills

The place of ryegrass on the leached soils is taken by browntop, Chewings fescue, and *Notodanthonia clavata* (figure 26). These species characterise pastures of this type. Ryegrass sometimes makes an appearance in moist places around sheep camps but there is every indication that it is excluded by its requirement for soil of higher fertility. White clover, on the other hand, is nearly always present, though often in a very depauperate form.

The interesting botanical feature of the pastures is their diversity of both native and introduced species. Typical areas contain browntop, Chewings fescue,

and *Notodanthonia clavata* as the dominant grasses but crested dogstail, sweet vernal, white clover, suckling clover, catsear, hawkbit, selfheal, *Centella*, *Luzula congesta*, *Dichondra repens*, *Hydrocotyle moschata*, *H. americana*, *Microtis unifolia*, and marsh thistle are usually there also. Among the plants found in these pastures and seldom if ever elsewhere are purging flax, *Myosotis discolor*, *Potentilla anglica*, and *Nertera setulosa*.

Where rotting logs remain *Helichrysum filicaule*, *Acaena novae-zelandiae*, *A. anserinifolia*, and *Notodanthonia gracilis* are plentiful. In patches of turf destroyed by the frequent and severe attacks by grass grubs the vegetation soon disintegrates and disperses. The first colonist is *Notodanthonia gracilis*. At the same time *Helichrysum filicaule*, freed from its competition from browntop, increases in density. Eventually browntop re-establishes and restricts other species until the next grass grub attack wipes it out.

In the permanently moist places grasses give way to *Juncus articulatus*, creeping buttercup, and other herbs, among them *Eleocharis gracilis*, *Centella*, selfheal, and *Haloragis micrantha*. The sites that are wetter favour kneed foxtail but it does not become abundant unless there is some denudation through soil disturbance.

On the infertile terraces near Aokautere there are some patches of turf of unusual composition. *Schoenus apogon* and Chewings fescue are frequent wherever there is manuka in this district, but in a few places these make a close sward with *Microlaena stipoides*, sweet vernal, hawkbit, and *Carex flacca*. This is not the best kind of sward from a farmer's point of view. It represents the last stage in pasture deterioration before manuka takes over.

Lawns and playing fields

The composition of turf varies with the soils it grows on, the seed mixture used, the age of the turf, and the treatment it has received. The model browntop and Chewings fescue sward remains an ideal that is seldom achieved. A major reason for failure is that only expensive seed mixtures have these components in sufficient quantity to make a good lawn. Very often treatment after sowing does not always encourage browntop and fescue. Occasionally a good browntop or good fescue lawn is seen but the two species seldom combine to make a fine turf. It seems that fescue can too easily be eliminated by the closeness of mowing that browntop can tolerate, the fescue persisting in places not regularly reached by the mower. Sometimes fescue will be prominent in a lawn that does not receive persistent close mowing.

In some instances weeds are allowed to suppress the seedling grasses. The main competitors in autumn-sown lawns are the annuals—spurrey, chickweed, *Veronica persica*, *Poa annua*, and sometimes wild turnip. In older lawns the more aggressive weeds are mostly perennials—pennyroyal, creeping buttercup,

Hydrocotyle moschata, *Thuidium* moss, white clover, perennial ryegrass, Yorkshire fog, crested dogtail, sweet vernal, and the flatweeds, particularly *Bellis perennis*, catsear, and hawkbit.

The average Manawatu lawn, if well kept, is attractive throughout most of the year but not because of its purity. It is a minor botanical garden. There is seldom any difficulty in finding 20 different species in the turf. Species other than those already mentioned which occur regularly are suckling clover, lesser suckling clover, annual mouse-ear chickweed, *Veronica serpyllifolia*, dandelion, hawksbeard, fiddle dock, *Dichondra repens*, narrow-leaved plantain, pearlwort, *Oxalis exilis*, yarrow, *Gnaphalium spicatum*, selfheal, *Poa trivialis*, and crested dogtail.

In lawns which are mown closely Onehunga weed is usually troublesome. Some good examples of the encouragement of this weed are seen on footpath lawns. The well-kept lawn is usually infested while the unloved lawn next door is free of it.

Weeds which occur with a fairly low frequency are paspalum, *Veronica arvensis*, Scotch thistle, *Gnaphalium involucreatum*, *Bromus breviaristatus*, *Hypericum japonicum*, *Pratia angulata*, *Nertera setulosa*, *Notodanthonia racemosa*, broad-leaved dock, and musky storksbill. *Hydrocotyle americana* and allseed occur frequently on shaded lawns.

Lawns on sand country often consist largely of browntop with varying amounts of *Microlaena stipoides*, *Notodanthonia racemosa*, annual grasses, and annual legumes. In some places near the coast buffalo grass has been encouraged. At Foxton Beach buck's-horn plantain and strawberry clover grow in some lawns. Indian doab is not abundant on sand country but has become more widespread since I have been working in the Manawatu.

Grasslands of the water reserves

On the summit of Arawaru at 767 m there is a minor stand of *Chionochloa conspicua* and *C. rubra* × *flavescens* that has the appearance of being natural. This assumption has become more difficult to substantiate in the light of the vegetation changes which are known to have occurred in the region. The death of *Olearia colensoi* or its removal to expose the trig beacon may account for the few hundred square metres of grassland. It is said that the Maoris sent smoke signals from Arawaru so it is possible that the summit was cleared long ago.

Grasslands dominated by native species are conspicuous features of the Tiritea and Kahuterawa catchments. Four tall species are present—two species of toetoe (*Cortaderia toetoe* and *C. fulvida*) and two species of tussock (*Chionochloa cheesemanii* and *C. conspicua*) (figure 30). The interesting



Figure 30 A stand of *Chionochloa cheesemanii*, one of the four tall native grasses which occupy land once covered in forest in the Tiritea reserve. 1964

feature is the relatively little overlap in the distribution of the species. Some pure stands cover several hundred square metres and are apparently quite youthful. Any one site could favour each species more or less equally except that the tussocks tend to grow on steeper slopes more readily than the toetoes. The recent development of these patches of grassland is borne out by the observations of F. G. Hayes who noted very little tussock and toetoe in 1927 in places where they are now abundant. Another line of evidence is from the logs on these grassy patches. It is apparent that, when the primary forest canopy opened a few decades ago, the tussocks which were minor elements of the forest spread extensively. The toetoe species were obviously able to move from the stream margins and slips which they occupied earlier. They occur at all altitudes but the tussocks grow only above 395 m.

VEGETATION OF THE UNSTABLE SAND COUNTRY

In the distant past there were two periods when sand drifted freely, each followed by an interval of stability which allowed vegetation to establish. The past and present plant cover of these stable dunes and plains has been considered in this handbook under the appropriate vegetation headings. A coastal strip where sand is still drifting or has been stable for no more than 50 years is so distinctive that it warrants separate treatment. The environment

and topography are considered in some detail because a knowledge of these gives meaning to the seemingly haphazard distribution of the plant communities.

This piece of sand country runs the full length of the Manawatu coast and extends inland for varying distances up to 3 km where the drifting dunes, in many places, meet a chain of lagoons (*map 2*).

Environment and topography

The forces at work shaping the sand country can be seen clearly near the coast. Except for two brief papers (Esler 1969b, 1970) the interplay of sand, plants, moisture, wind, and topography have not been recorded in much detail in New Zealand. Nowhere else known to me can this interplay be seen so clearly. Inevitably stabilisation of drifting sand followed by invasion by tree lupin, tall fescue, and other exotic plants will soon alter and annihilate these unique stands. It is fortunate that they have persisted long enough to be described. Such communities must have occurred in other parts of New Zealand, but have apparently been destroyed. For these reasons I dwell at some length on this aspect of sand country ecology.

It seems a paradox that in this harsh unstable coastal environment native plants predominate over exotics on the sand plains. The exotics are versatile and diverse in species but, lacking the rhizomatous habit and tolerance of prolonged inundation, they are no match for a small group of resident native herbs which possess these features.

Dry sand is constantly on the move covering, uncovering, and abrading plants as it is propelled along by the strong and persistent north to north-west winds. The winds also carry large amounts of salt and have strong desiccating effects. The rapidly changing temperatures and high reflectivity of the sand are further impediments to plant growth. Added to this the sand is very free-draining and relatively infertile.

Sand does not move in a haphazard manner. In a strong wind sand grains become airborne, strike the surface at intervals and rebound back into the air. The movement continues as long as the propelling force persists and the sand remains dry. The presence of moist sand lying just above the water table determines the lower limit to which sand can be removed (deflated). The water table under the plains slopes evenly at about 2 m per kilometre and forms a base line for deflation—one factor which imposes order on this complex and seemingly disorderly topography.

Deposition of sand follows regular patterns set by the laws of aerodynamics and the angle of rest of the sand. Sand moving across the bare surfaces is deposited in mounds with some consistency of form (*figure 31*). The windward slope (termed the wind sweep) is long and even and acts as a ramp. In the lee of the mound sand loses its momentum and tumbles down the slip face.

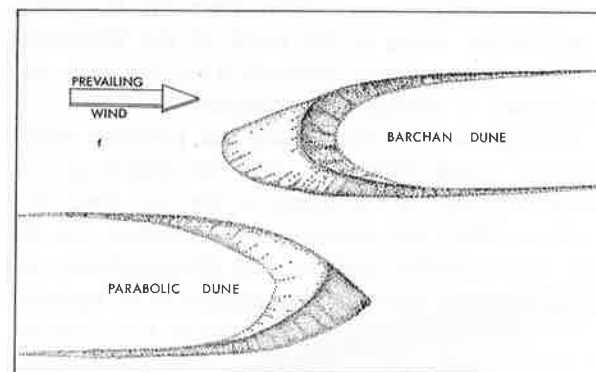


Figure 31 Two forms of moving dunes. Barchan dunes develop to a very limited extent where vegetation is sparse or absent. Parabolic dunes are a feature of the Manawatu sand country. A light cover of vegetation causes the arms to trail behind the main body of the moving sand.

In plan, such a mound or dune is crescent-shaped with arms extending forward. These barchan dunes are typical of dry sandy deserts and are developed to only a very minor extent in the Manawatu. Where some vegetation is present the dune has more restricted mobility and the arms are directed backward and trail behind the moving dune. Usually the arms elongate and a parabolic dune is formed. In the Manawatu dunes of this nature form an extensive complex series, much of the landscape being made up of the broad plains between the trailing arms which in some places are many kilometres long. Before dunes of this nature finally stabilise the wind sweep steepens, probably because of diminishing supply of sand.

The development of the present coastal topography can be traced from aerial photographs taken over a period of time. A phase of active sand movement began more than a century ago, most probably as a result of heavy grazing by cattle. Trampling coupled with the elimination of spinifex, the major sand binding plant, set much sand on the move. It drifted about 3.2 km inland, enveloping farmland, threatening the (now lifted) Palmerston North to Foxton railway line, and altering the size, shape, and position of lagoons on the landward margin of the drifting sand. The 1942 aerial photographs show no fore dune and only limited plains but extensive areas of low sandy ridges running inland towards the east-south-east. It seems that, at this stage, the re-establishment of spinifex and the introduction of marram, a grass of European origin, had not produced a significant influence on the topography. Later photographs show the formation of the fore dune, the parabolic dunes, and the extensive plains.

The fore dune

The present fore dune is well defined and fairly stable except for a few breaches. Tremendous quantities of sand are deposited on the beach from a

drift moving down the coast carrying erosion debris from the Rangitikei, Whangaehu, Wanganui, and minor rivers to the north of the Manawatu. The sand is so plentiful that the coast moves seawards 0.6–1.0 m each year and the fore dune moves forward a corresponding distance.

The large quantity of drifting sand, the very strong and persistent winds, erosion caused by the extreme spring high tides, and the drenching with salt spray impose very severe hardships on the plants of the fore dune, thus limiting the number of species which can survive in this situation. On the seaward face of the dune only spinifex, marram, and *Desmoschoenus* are prominent. Spinifex is by far the most important and appears to be replacing marram which has proved to be inferior as a sand binder in this situation. *Desmoschoenus* is of fairly minor importance.

The three primary sand binders impose their influence on the form of the fore dune where each is dominant (figures 32 and 33). Each species has a different form influencing patterns of wind flow. Thus each builds its own form of dune.

Spinifex has fast-growing, extensive rhizomes and a diffuse, flexible cover of leaves. Each blade and tuft of grass calms the wind, and sand is deposited behind it. Because rhizomes grow down slopes and into hollows, dunes covered with spinifex are normally smooth and regular. Salt water does not adversely affect the rhizomes which reach right down to the base of the fore dune where they prevent wave erosion. As more sand accumulates on the face of the fore dune more rhizomes are produced to permeate and stabilise it. A section of fore dune dominated by spinifex is regularly about 6 m high and slopes evenly at 14–16° from the horizontal.



Figure 32 Fore dune on the Manawatu coast. The form of the dune depends largely on the species of sand binding plant growing on it. Contrast the smooth, even spinifex dune in the foreground and distance with the steeper irregular marram dune (right centre) and the low *Desmoschoenus* dune (centre). See figure 33. 1969

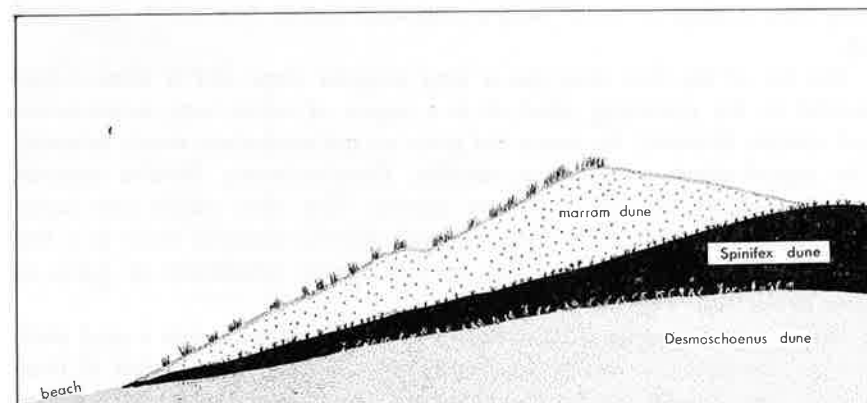


Figure 33 Profile of marram, spinifex, and *Desmoschoenus* fore dunes on the Manawatu coast.

Marram takes second place to spinifex on the fore dune. It does not tolerate salt water as well as spinifex, does not spread as rapidly, and tends more to form clumps. The dune that it forms is stepped and steep on the seaward face, reaches a maximum height of about 8 m and has a slope of 24–28°. Although it thrives in accumulating sand, marram does not compete well with spinifex, nor does it hold sand as well. Lacking the diffuse, open habit of spinifex, marram causes winds to deposit sand heavily in the lee of clumps. Wind then funnels between the clumps and the mounds causing channels which the very extensive root systems cannot always prevent from developing. Thus marram dunes are nearly always eroding dunes with blow-outs undermining the marram, which remains for a time holding a turret of sand but under these deflating conditions soon succumbs.

The failure of marram in this situation does not condemn its value in less severe regimes, or belittle its usefulness for artificial stabilisation. When marram is planted to stabilise sand on a beach, spinifex either establishes naturally from seed or is hand sown to form the ultimate cover.

Desmoschoenus is well adjusted to the fore dune habitat but is less abundant in this situation than spinifex or marram. Like spinifex it tolerates salt water fairly well. It also grows down slopes and prevents undercutting by the wind, but to a lesser degree. Once deflation is under way *Desmoschoenus* is powerless to stop it. The species thrives best where sand is accumulating. Again, the form of the plant determines the form of the dune it builds. *Desmoschoenus* has long rhizomes sending up fairly open tufts of leaves at intervals. This gives a diffuse cover that promotes the formation of smooth dunes but, because the leaves are fairly inflexible and the rhizomes elongate slowly, sand does not accumulate freely. *Desmoschoenus* fore dunes seldom exceed 3 m in height.

They have a slope of 8–14° with a somewhat convex face and a fairly level top.

The lee of the fore dune has a long irregular slope and is often ridged parallel to the prevailing wind. It is a region of active sand accumulation and erosion. However, the losses and gains are not everywhere evenly balanced. The typical plants are spinifex, marram, *Desmoschoenus*, *Pimelea arenaria*, *Cassinia leptophylla*, and *Coprosma acerosa*. The alien plants tree lupin, catsear, haretail, *Erigeron floribundus*, and *Bromus diandrus* occur in a few stable parts of the fore dune but are not regular inhabitants of dunes so close to the sea.

The landward margin of the fore dune tails off gradually into a sand plain or ends abruptly at a wet hollow about 50 m from the beach. Most of these hollows have *Selliera* and *Carex pumila* growing in them and contain midden material. In many there are also totara logs in an advanced stage of decay. Presumably they were stranded on the beach more than a century ago, were covered by the advancing fore dune, and have been uncovered again in the rear of the dune.

The sand plains

The sand plains present a complex picture unless considered in relation to their origin (figure 34). In its simplest form a plain is the area between the trailing arms of a parabolic dune. It is usually deflated over part of its area to the moist sand lying a little above the water table. As the parabolic dune advances each plain increases in size. Eventually the trailing arms break away from the fore dune and advance inland, often several hundred metres in a dry summer. The arm, now taking the form of a longitudinal dune, is usually common to two parabolic dunes. As it retreats inland from the fore dune two adjacent plains become one. When several such plains coalesce a large basin is formed. This is bounded on the seaward margin by the fore dune and on the landward margin by the apices of several parabolic dunes forming what can best be regarded as a rear dune.

Within the basin small plains develop in a sequence in the wake of the moving rear dune (figure 35). The youngest are close to the rear dune. Here sand is deflated to near the water table in summer and a wet hollow develops. In the wettest parts there soon appear a set of small creeping native plants—*Myriophyllum votschii*, *Limosella lineata*, *Eleocharis neozelandica*, *Ranunculus acaulis*, and *Selliera*. Figure 36 illustrates the influence of position of the water table on the distribution of these and other plants. Around the edges *Carex pumila* forms a continuous fringe. As it spreads it gathers sand to form a level mound about 0.3 m higher than the depression. Seeds of marram and *Desmoschoenus* germinate on the moist sand on its margin and soon come to occupy the mound and gather more sand about them. While this

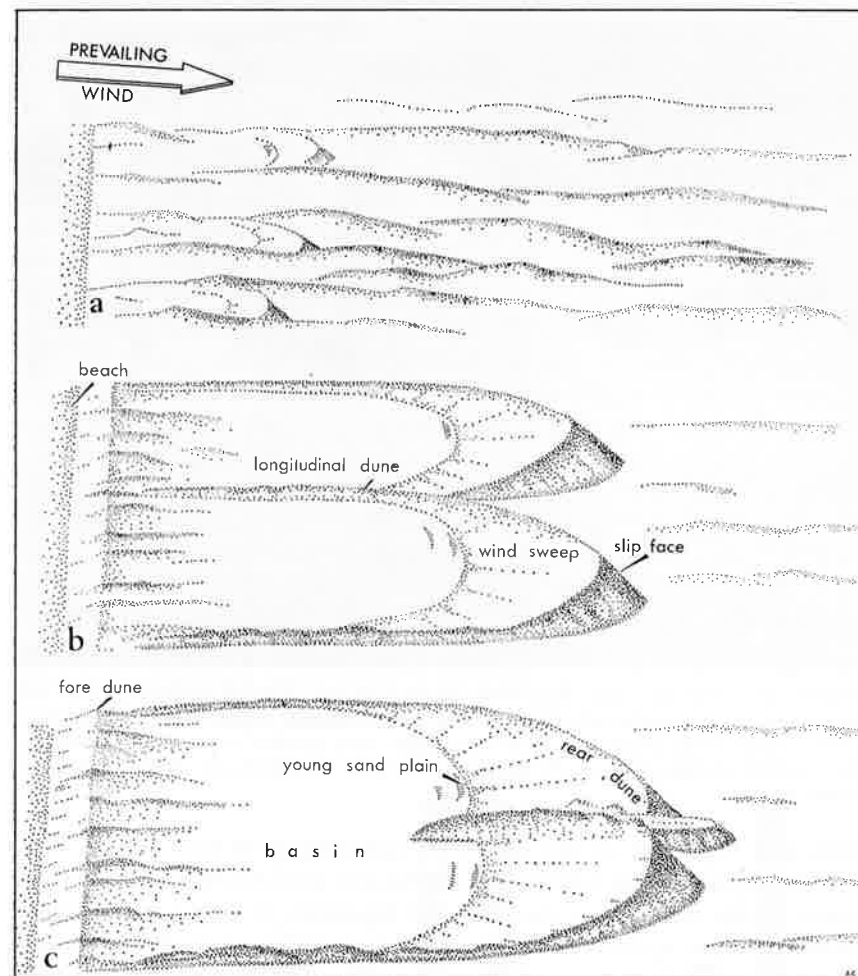


Figure 34 The evolution of Manawatu coastal topography. a. Aerial photographs taken in 1942 of the coast between Foxton Beach and Himatangi Beach show long, low, irregular dunes and no distinct fore dune. b. Later a fore dune developed and the minor sand plains behind it enlarged with the creation of parabolic dunes. c. A rear dune and large basin developed as adjoining parabolic dunes coalesced.

is developing the rear dune has moved another 20 m or so away and a new depression has developed in its place.

As the new plain is becoming colonised the vegetation in the previous plain develops further. Usually *Leptocarpus* becomes dominant and is joined by *Schoenus nitens* and other plants. *Cortaderia toetoe*, pampas, and cabbage trees appear later, sometimes also manuka and *Olearia solandri* in small quantities.



Figure 35 A young, wet sand plain formed on the seaward margin of the rear dune shown in part in the left distance. The *Desmoschoenus* (right centre) and marram in the distance have established from seed. They have invaded a low dune formed by *Carex pumila* (foreground) on the margin of an older sand plain out of the photograph on the right. 1969

Plains with vegetation of this nature are approaching 50 years of age.

Some wet plains gather further sand and become quite dry. Where *Leptocarpus* persists in this habitat it is clear that the deposition of sand is fairly recent. Normally *Scirpus nodosus* takes over from *Leptocarpus* as conditions become drier. It occupies dry plains and low dunes as well, particularly near the fore dune. This seaward margin of the basin is usually a complex region. The oldest of the plains are often submerged in sand drifting off the fore dune and are replaced by low irregular dunes covered with marram, *Desmoschoenus*, *Scirpus nodosus*, tree lupin, *Oenothera stricta*, catsear, hawkbit, annual grasses, and annual legumes. For some reason as yet not fully explained spinifex does not usually extend more than a few hundred metres inland from the beach. In this same region there is often a stand of *Carex pumila* forming a low level dune on the margin of a wet plain. This occurs where sand is swept away rapidly from the rear margin of the foredune and a wet hollow has developed.

A sand basin of a similar nature is described in Esler (1969b). In that example sand was moving freely over most of the basin. Sand plains formed

at the foot of the wind sweep and along the margins of the arms of the parabolic dunes. Older plains became obliterated by moving sand.

The rear dune

The rear dune is made up of the apices of coalesced parabolic dunes. The wind sweep is long, wide, and gently sloping. In places it forms channels around turrets capped with marram, *Desmoschoenus*, or dune shrubs. The sand is pushed over the slip face lying at an angle of 36° either on a broad front or as a narrow tongue. On the lee, sand spills onto a plain or is lost among hillocks covered with marram and tree lupin. Where marram is present (usually because of planting) sand movement is halted, the wind sweep steepens and the dune becomes stabilised under a cover of marram, tree lupin, and *Muehlenbeckia complexa*.

Stabilised dunes of this form lie more or less parallel to the coast and are modern features because the native flora lacked species capable of halting moving sand in this manner. In the pre-marram era the rear dune, if it formed at all, would have been a stage in the union of many smaller dunes into one large parabolic dune. This would have continued migration until the supply of sand, now spread over many hectares, was insufficient to maintain its movement. Many dunes of this nature are preserved in the landscape of the older parts of the Manawatu sand country.

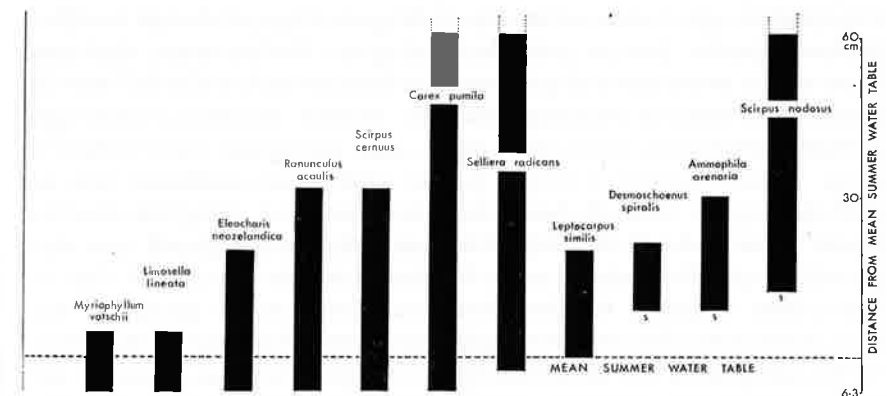


Figure 36 Histogram showing vertical distribution of plants on sand plains relative to the water table. The information was obtained by measuring the range of elevations at which species occurred. With the water level occasionally 30 cm higher in winter plants on the lower sites were submerged at times, and some of the dune plants (*Desmoschoenus*, *Ammophila*, and *Scirpus nodosus*) for shorter periods. Only seedlings of these were recorded. Data from Esler (1969b).